

## Project Goal

Athletes, athletic trainers and coaches require a reliable, accurate device to record the possibility and severity of a head impact in real-time without hindering athletic performance.

## Background

Concussions are a major issue in contact sports, often going unreported due to subjective symptoms and pressure to continue play.

Traditional detection methods rely on self-reporting symptoms or visual observation by coaches or trainers, which are prone to error.

Current devices, such as mouthguards and helmets are sport-specific, bulky and can produce unreliable or inconsistent results to measure impacts or injury potentials.

Another neck collar device applies vascular pressure for brain protection, but doesn't measure impacts.

The **Impact360 collar** overcomes these limitations with a lightweight, ergonomic design that would encourage athletes in all sports to participate in protecting their own health.

## Device Design

### Mechanical Prototypes

- Impact 360 works by applying two sensors at the C5 and C2 vertebrae.
- 3D printed designs with different materials were tested, but transitioned to silicone after these proved uncomfortable and were not flexible enough for neck use.
- Refined the silicone mold designs in SolidWorks then each mold was 3D printed in 3 parts (top, bottom, hinge).
- Produced casts with Smooth-On Ecoflex™ 00-30 and silicone polyurethane.
- Silicone was selected for its comfort, elasticity and water resistance.

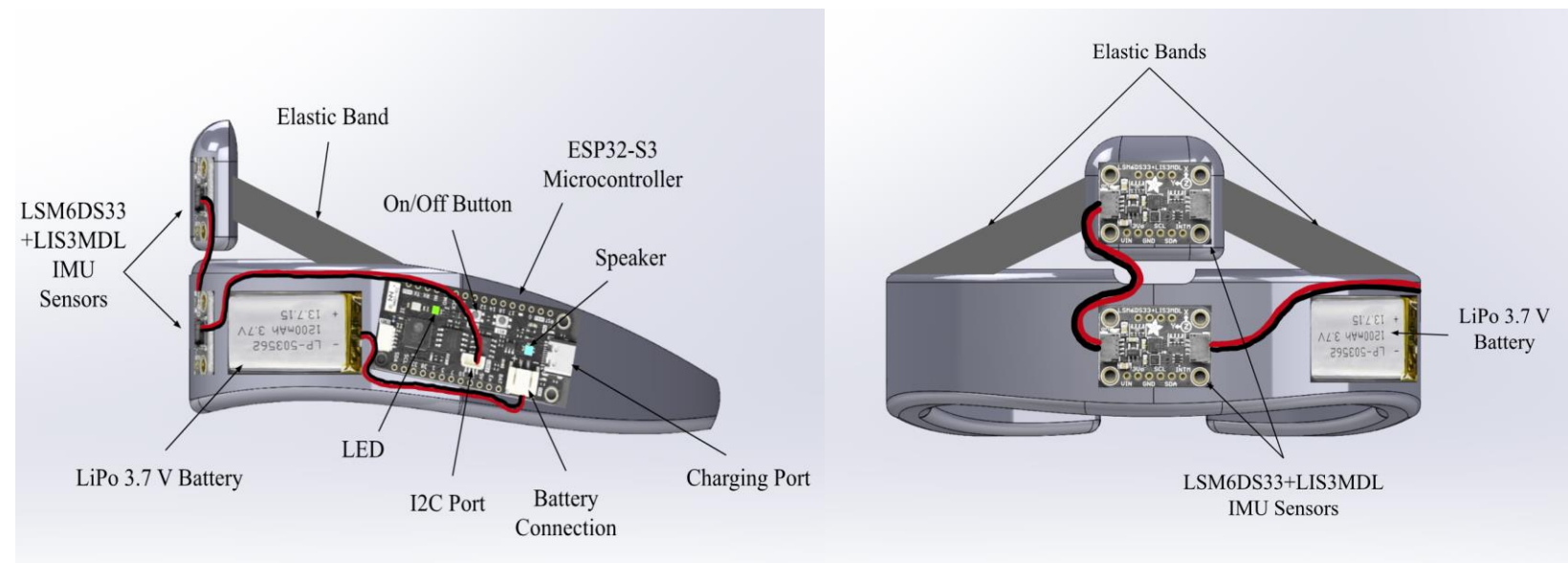


Fig 3. Impact360 device mechanical design and electronics layout.

### Electronic Prototypes

- Dual IMU sensors (LSM6DS33 + LIS3MDL) positioned at C2 and C5 detect differential motion.
- ESP32-S3 microcontroller processes sensor data and transmits impact alerts via Bluetooth while powered by a 3.7V LiPo battery.
- The sensors reliably captured all motions, confirming its suitability for detecting rotational and linear impacts.
- Impact severity thresholds were from car whiplash testing and head impact sensor data from O'Connor 2017 and Smotrova 2021.
- A spike in linear acceleration or angular velocity triggers an 80 ms data window and maximum sustained values were used to classify severity.

```
def classify_severity(total_lin, total_ang):  
    if total_lin > 80 or total_ang > 80:  
        return "RED"  
    elif total_lin > 60 or total_ang > 45:  
        return "YELLOW"  
    elif total_lin > 40 or total_ang > 25:  
        return "GREEN"  
    else:  
        return "NOT SUSTAINED"
```

Fig 5. CircuitPython code impact severity classification logic.



Fig 1. Final prototype design, complete with sensor placement, silicon exterior and stable C2 joint.

### Functional Prototypes

- A calibration factor was derived by comparing sensor data to reference values to align signals across axes.

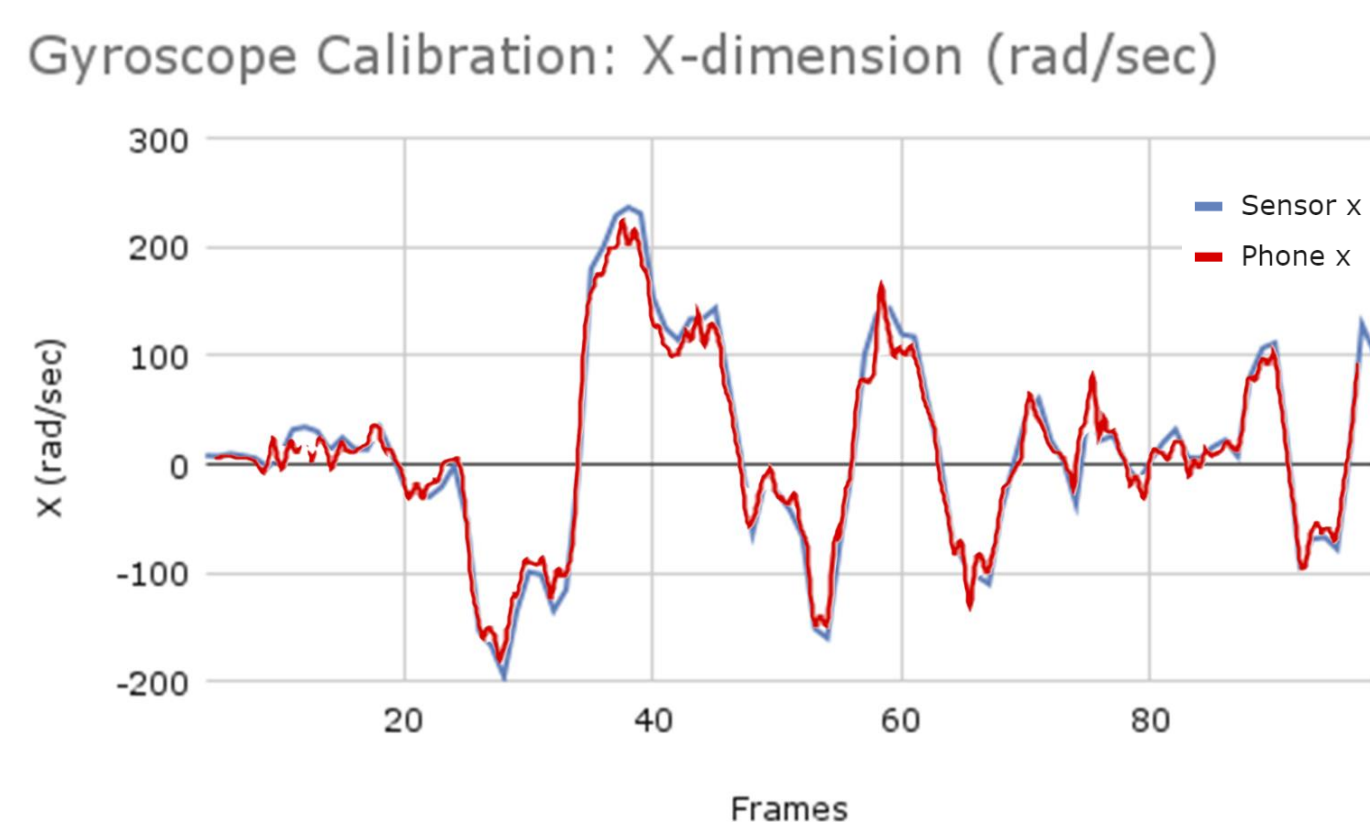


Fig 2. Comparison of gyroscope data from Phyphox app and LSM6DS33 sensor.

- Sensor placement demonstration of the ability to collect meaningful data.

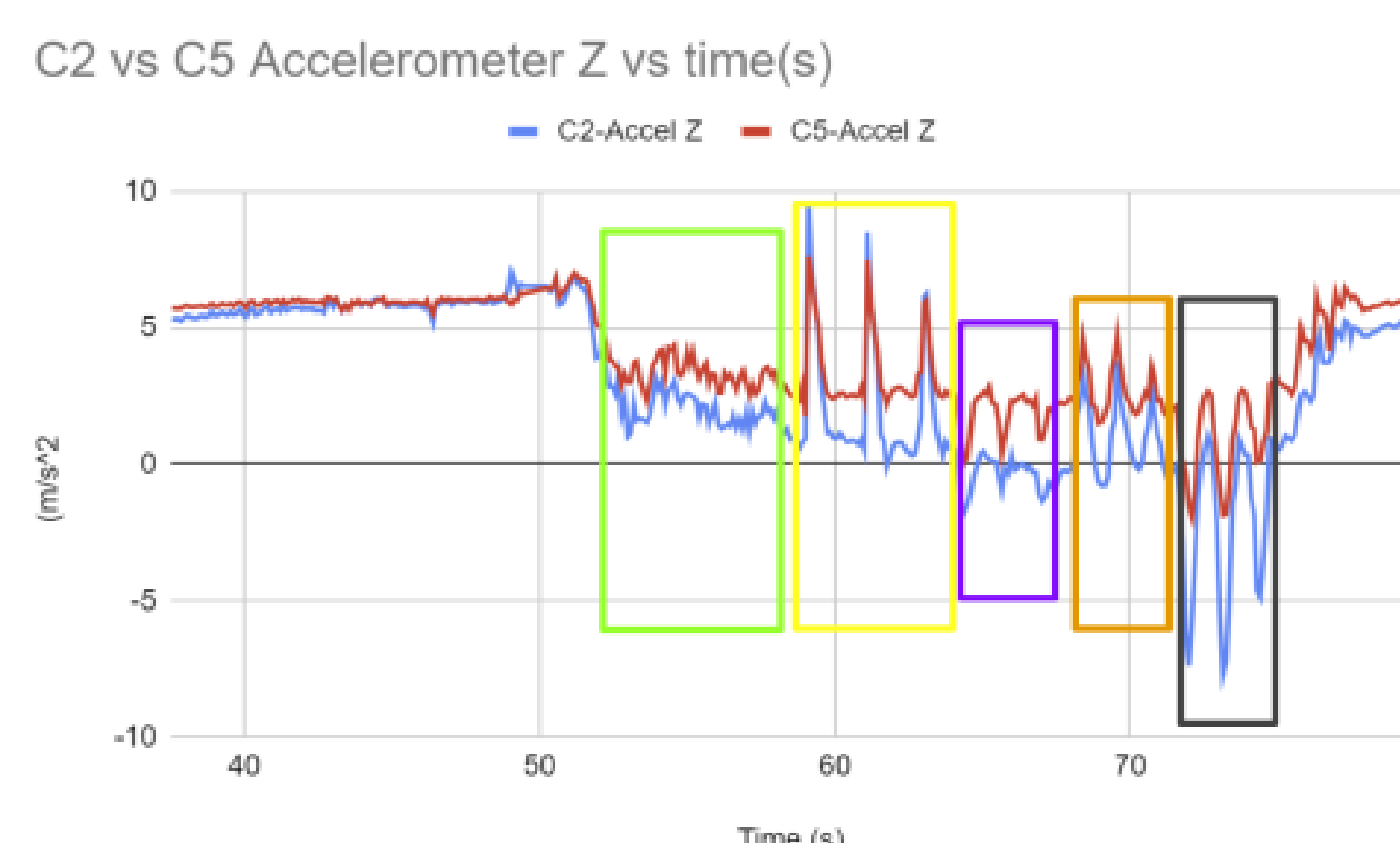


Fig 4. Data verification displaying different forms of motion; walking (green), neck flexion (yellow), right head rotation (purple), left head rotation (orange) and neck extension (black).

- App displays real-time impact detections for timestamp, linear acceleration (g), angular velocity (rad/s) and corresponding severity classification (Not Sustained, Green, Yellow, Red).

```
14:28:29 RX IMPACT DETECTED  
14:28:30 RX Impact @ 238.59s |  
14:28:30 RX NOT SUSTAINED  
14:28:30 RX Linear:  
14:28:30 RX r: 36.66g | Angular:  
14:28:30 RX 7.16rad/s  
14:28:30 RX Total: GR  
14:28:30 RX een=0, Yellow=0, Red  
14:28:30 RX =0  
14:28:35 RX IMPACT DETECTED  
14:28:35 RX Impact @ 244.03s | G  
14:28:35 RX GREEN  
14:28:35 RX Linear: 42.10g  
14:28:35 RX Angular: 7.54rad/s  
14:28:35 RX Total: Green=1, Yel  
14:28:35 RX low=0, Red=0
```

Fig 6. Impact detection output from Impact360.

## Testing Procedures

### Impacts

- Simulated whiplash forces using a suction punching bag on a spring with sensors placed at four marked positions to represent impacts from the front of the head as well as side to side.
- Tested at consistent angles and recorded impact detection to ensure consistent and reliable results.



Fig 7. Impact testing set up with punching bag and sensor placement.

### Participants (with IRB approval)

- Participants were fitted with Impact360 collar and asked to do a range of natural athletic movements.
- These were monitored to determine if normal movements would set off false positives.
- The number of impacts was also compared to data collected when the sensors were manually adhered to C2 and C5 to determine if the back piece moves with the participants head as impacts occur.



Fig 8. Final design placement during participant testing.

### Comfort (with IRB approval)

- Participants were asked to wear the collar for different segments of time, (15 minutes, 30 minutes, one hour) and rate the comfort and ease of wear.

### Athletic Performance (with IRB approval)

- Participants were fitted with Impact360 collar and completed team or individual practices to monitor impacts in a realistic athletic setting.



Fig 9. Athletic performance testing on LTU women's lacrosse team.

## Results

- Accurately detected impacts 10 out of 10 times for each angle and sensor placement.
- Showed consistent sensor output and impact detection given consistent input force.
- 21 out of 21 participants successfully completed participant testing without triggering an impact alert.
- 21 out of 21 participants found that the collar did not cause significant or moderate interference with athletic activity.
- 75% of participants who described the collar as uncomfortable measured a neck circumference over 39 cm, indicating a sizing issue.
- Throughout a 30 minute practice period, Impact360 detected only one "Not Sustained" impact, which was expected.

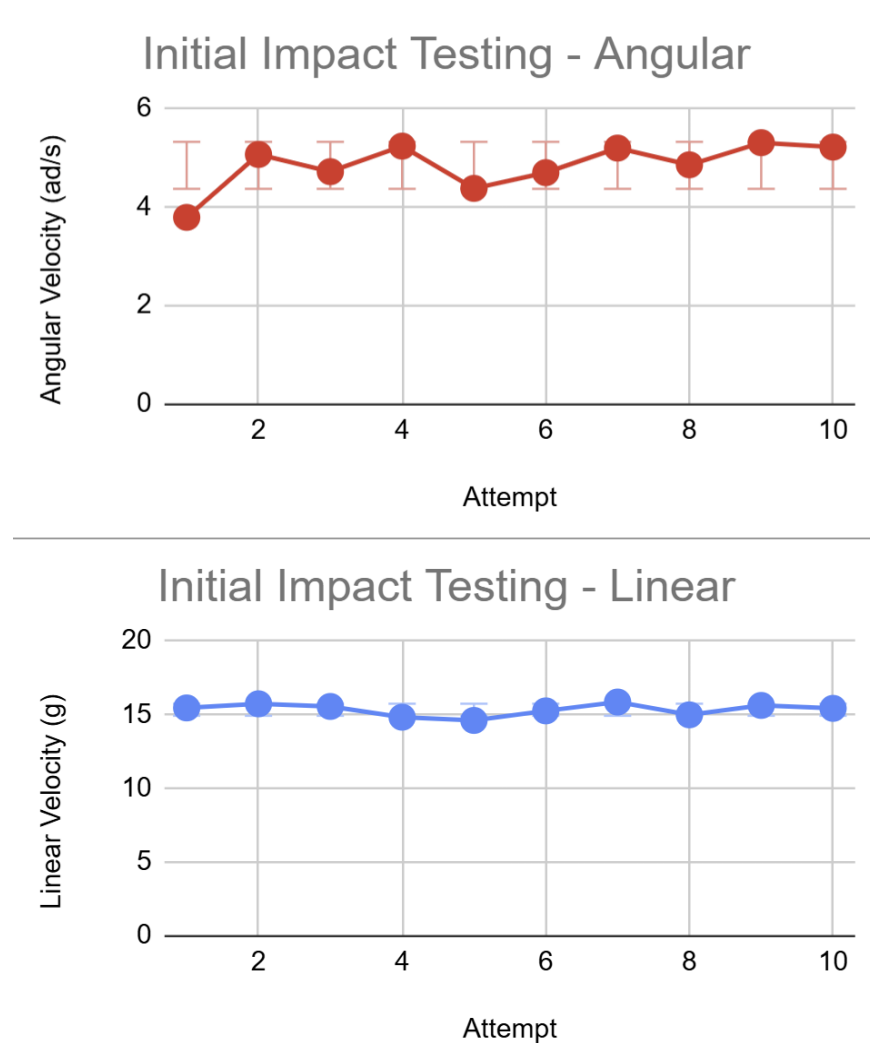


Fig 10. Initial impact testing results to demonstrate consistency in sensor readings.

## Conclusion and Future Work

- Impact 360 can accurately detect severity of head impacts based upon research thresholds and user testing.
- Participants found the collar to have minimal to no interference within athletic performance testing.
- The device poses low risk when worn around the neck, with only mild discomfort occasionally noted during athletic activity.

**Future work** will focus on developing adjustable sizing or *multiple sizes* to accommodate various neck dimensions, designing a mobile application for real-time impact monitoring by athletic trainers, and integrating the electronic components more seamlessly within the collar structure. Additionally, *crash test dummy experiments* will be conducted to further validate and refine the impact detection thresholds.

## Acknowledgements

Prof. John Peponis, Andrew Ulaszek, Department of Biomedical Engineering Faculty & Staff, Lawrence Technological University  
Noah Pultorak, Biomedical Engineering Technical Advisor  
Student volunteers for device testing, Lawrence Technological University

## References

- Cobb, B. (2013). Measuring Head Impact Exposure and Mild Traumatic Brain Injury in Humans. In *University Libraries at Virginia Tech*. Virginia Tech – Wake Forest University, Center for Injury Biomechanics.
- O'Connor, K. L., Rowson, S., Duma, S. M., & Broglio, S. P. (2017). Head-Impact–Measurement Devices: A Systematic review. *Journal of Athletic Training*, 52(3), 206–227.
- Smotrova, E., Morris, L., & McNally, D. (2021). Comparison of standard automotive industry injury predictors and actual injury sustained during significant whiplash events. *European Spine Journal*, 30(10), 3043–3058.